

CHAPTER 5: DEADLOCK

Exercises:

ANSWERS

1. Consider the matrix tables below and answer the following questions:

Claim matrix C			
	R ₁	R ₂	R ₃
P ₀	4	4	3
P ₁	7	2	1
P ₂	5	4	2
P ₃	5	0	2
P ₄	4	3	3

Allocation matrix A			
	R ₁	R ₂	R ₃
P ₀	0	0	2
P ₁	2	0	0
P ₂	2	1	1
P ₃	3	0	2
P ₄	0	0	2

Available vector V			
	R ₁	R ₂	R ₃
	3	3	2

Answer the following questions using the banker's algorithm.

- What is the content of the matrix Need?
- Is the system in a safe state? If yes, what is the amount of each resource? Show all your steps.

Answers

a)

Need C-A			
	R ₁	R ₂	R ₃
P ₀	4	4	1
P ₁	5	2	1
P ₂	3	3	1
P ₃	2	0	0
P ₄	4	3	1

- b) Give available resource to P₂, V = (0, 0, 1)
P₂ release resources, V = (5, 4, 3)

Give available resource to P₁, V = (0, 2, 2)
P₁ release resources, V = (7, 4, 3)

Give available resource to P₃, V = (5, 4, 3)
P₃ release resources, V = (10, 4, 5)

Give available resource to P₄, V = (6, 1, 4)
P₄ release resources, V = (10, 4, 7)

Give available resource to P₀, V = (6, 0, 6)
P₀ release resources, V = (10, 4, 9)

System is not in a safe state.

2. Consider the matrix tables below and answer the following questions:

Request matrix Q					Allocation matrix A					Available vector V				
	R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄
P ₀	1	1	0	0	P ₀	0	1	2	0		2	1	0	0
P ₁	2	0	0	1	P ₁	0	0	1	0					
P ₂	1	0	1	0	P ₂	2	0	0	1					
P ₃	3	0	3	0	P ₃	1	1	1	1					
P ₄	2	4	1	1	P ₄	0	1	2	2					

Apply the steps of **deadlock detection** to the data above.

Answers

$$W = V = (2 \ 1 \ 0 \ 0)$$

$$\text{Mark } P_0, W = (2, 1, 0, 0) + (0, 1, 2, 0) = (2, 2, 2, 0)$$

$$\text{Mark } P_2, W = (2, 2, 2, 0) + (2, 0, 0, 1) = (4, 2, 2, 1)$$

$$\text{Mark } P_1, W = (4, 2, 2, 1) + (0, 0, 1, 0) = (4, 2, 3, 1)$$

$$\text{Mark } P_3, W = (4, 2, 3, 1) + (1, 1, 1, 1) = (5, 3, 4, 2)$$

Cannot Mark P₄, → deadlock.

3. Consider the matrix below and answer the following questions:

Pertimbangkan matriks di bawah dan jawab soalan yang berikutnya:

Process	Claim Matrix, C			Process	Allocation Matrix, A		
	R1	R2	R3		R1	R2	R3
A	9	10	11	A	3	4	3
B	6	7	8	B	2	3	2
C	3	4	5	C	1	2	1
D	0	1	2	D	0	1	0

Process	Need Matrix C - A		
	R1	R2	R3
A	6	6	8
B	4	4	6
C	2	2	4
D	0	0	2

R1	R2	R3
11	11	11

Resource vector, R

R1	R2	R3
5	1	5

Available vector, V

a) Using Banker's algorithm, complete the following table by stating the process that would be executed in order and the available vector after each process runs to completion.

Menggunakan algoritma Banker, lengkapkan jadual berikut dengan menyatakan proses yang akan dilaksanakan mengikut turutan dan vektor "available" selepas setiap proses selesai perlaksanaannya.

Answer:

Process to be executed (in order)	Available vector, V		
	R1	R2	R3

1st process:	D	5	2	5
2nd process:	C	6	4	6
3rd process:	B	8	7	8
4th process:	A	11	11	11

b) Referring to your answer in (a), is any deadlock detected? Why?

Merujuk kepada jawapan anda di (a), adakah kebuntuan dikesan? Kenapa?

No, deadlock is not detected.

Because all processes can be executed with enough number of resources.

4. Consider the matrix tables below and answer the following questions:

Pertimbangkan rajah matrik di bawah dan jawab soalan-soalan berikut:

Request matrix Q					Allocation matrix A					Available vector V				
	R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄
P ₀	1	0	1	0	P ₀	0	1	2	0		4	2	0	0
P ₁	2	0	0	1	P ₁	0	0	1	0					
P ₂	1	1	0	0	P ₂	2	0	0	1					
P ₃	2	4	1	3	P ₃	0	1	2	2					
P ₄	3	0	3	0	P ₄	1	1	1	1					

Apply the steps of **deadlock detection** to the data above.

Jalankan langkah-langkah pengesanan kebuntuan kepada data di atas.

$$W = (4 \ 2 \ 0 \ 0)$$

$$\text{Mark } P_2, W = (4 \ 2 \ 0 \ 0) + (2 \ 0 \ 0 \ 1) = (6 \ 2 \ 0 \ 1)$$

$$\text{Mark } P_1, W = (6 \ 2 \ 0 \ 1) + (0 \ 0 \ 1 \ 0) = (6 \ 2 \ 1 \ 1)$$

$$\text{Mark } P_0, W = (6 \ 2 \ 1 \ 1) + (0 \ 1 \ 2 \ 0) = (6 \ 3 \ 3 \ 1)$$

$$\text{Mark } P_4, W = (6 \ 3 \ 3 \ 1) + (1 \ 1 \ 1 \ 1) = (7 \ 4 \ 4 \ 2)$$

Cannot Mark P₃, - deadlock.

5. Define deadlock. Is it possible to have a deadlock involving only one single process?

Explain your answer.

Takrifkan kebuntuan. Adakah berkemungkinan untuk mendapatkan kebuntuan yang melibatkan hanya satu proses tunggal? Jelaskan jawapan kamu.

Permanent blocking of a set of processes that either compete for system resources or communicate with each other.

No. This follows directly from the hold-and-wait condition.

6. What is deadlock avoidance? Give two approaches of deadlock avoidance.

Apakah ia pengelakan kebuntuan? Berikan dua pendekatan kepada pengelakan kebuntuan.

A decision is made dynamically whether the current resource allocation request will, if granted, potentially lead to a deadlock. It requires knowledge of future process request.

1. Do not start a process if its demands might lead to deadlock.
2. Do not grant an incremental resource request to a process if this allocation might lead to deadlock.

7. Consider the matrix tables below and answer the following questions:

Pertimbangkan rajah matrik di bawah dan jawab soalan-soalan berikut:

Request matrix Q					Allocation matrix A					Available vector V				
	R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄		R ₁	R ₂	R ₃	R ₄
P ₀	1	4	1	2	P ₀	0	1	2	0		5	2	1	1
P ₁	2	0	0	1	P ₁	0	0	1	0					
P ₂	1	1	2	0	P ₂	2	1	0	0					
P ₃	2	2	1	0	P ₃	0	1	2	2					
P ₄	1	5	0	4	P ₄	1	1	1	0					

Apply the steps of **deadlock detection** to the data above.

Jalankan langkah-langkah pengesanan kebuntuan kepada data di atas.

$$W = (5 \ 2 \ 1 \ 1)$$

$$\text{Mark } P_1, W = (5 \ 2 \ 1 \ 1) + (0 \ 0 \ 1 \ 0) = (5 \ 2 \ 2 \ 1)$$

$$\text{Mark } P_3, W = (5 \ 2 \ 2 \ 1) + (0 \ 1 \ 2 \ 2) = (5 \ 3 \ 4 \ 3)$$

$$\text{Mark } P_2, W = (5 \ 3 \ 4 \ 3) + (2 \ 1 \ 0 \ 0) = (7 \ 4 \ 4 \ 3)$$

$$\text{Mark } P_0, W = (7 \ 4 \ 4 \ 3) + (1 \ 1 \ 1 \ 0) = (8 \ 5 \ 5 \ 3)$$

Cannot Mark P₄, - deadlock.

8. There are three general approaches in dealing with deadlock, give and explain the approaches.

Terdapat tiga pendekatan untuk menangani kebuntuan, beri dan terangkan pendekatan tersebut.

- i. Deadlock Prevention - adopting a policy that eliminates one of the four conditions of deadlock (mutual exclusion, no-preemption, hold and wait and Circular wait)
- ii. Deadlock Avoidance - making the appropriate dynamic choices based on the current state of resource allocation
- iii. Deadlock Detection - detect the presence of deadlock and take action to recover.

9. List all four conditions when hold together can create a deadlock situation.

Senaraikan kesemua empat keadaan yang mana secara bersama boleh menyebabkan satu situasi kebuntuan.

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

10. Consider the matrix tables below and answer the following questions:

Pertimbangkan jadual-jadual matrik di bawah dan jawab soalan-soalan berikut:

Claim matrix C				Allocation matrix A				Need (C-A)				Available vector V			
	R ₁	R ₂	R ₃		R ₁	R ₂	R ₃		R ₁	R ₂	R ₃		R ₁	R ₂	R ₃
P ₁	4	3	3	P ₁	1	0	2	P ₁	3	3	1		2	3	2
P ₂	5	2	1	P ₂	2	0	0	P ₂	3	2	1				
P ₃	4	3	2	P ₃	2	1	2	P ₃	2	2	0				

Using Banker's algorithm, identify whether the system in safe state or not.

Menggunakan algoritma Banker's, kenal pasti sama ada sistem berada dalam keadaan selamat atau tidak.

Give available resource to P₃, A = 0 1 2
P₃ release resource, A = 4 4 4

Give available resource to P₁, A = 1 1 3
P₁ release resource, A = 5 4 6

Give available resource to P₂, A = 2 2 5
P₂ release resource, A = 7 4 6

System is in safe state

11. Consider the matrix tables below:

Pertimbangkan jadual-jadual matrik di bawah:

Request matrix Q				Allocation matrix A				Available vector V			
	R ₁	R ₂	R ₃		R ₁	R ₂	R ₃		R ₁	R ₂	R ₃
P ₁	2	0	2	P ₁	0	0	1		2	1	0
P ₂	2	1	1	P ₂	2	0	1				
P ₃	1	1	0	P ₃	0	1	1				
P ₄	0	2	4	P ₄	0	1	1				

Apply the steps of deadlock detection algorithm to the data above.

Gunakan langkah-langkah algoritma pengesahan kebuntuan terhadap data di atas.

$$W = (2 \ 1 \ 0)$$

$$\text{Mark } P_3, W = (2 \ 1 \ 0) + (0 \ 1 \ 1) = (2 \ 2 \ 1)$$

$$\text{Mark } P_2, W = (2 \ 2 \ 1) + (2 \ 0 \ 1) = (4 \ 2 \ 2)$$

$$\text{Mark } P_1, W = (4 \ 2 \ 2) + (0 \ 0 \ 1) = (4 \ 2 \ 3)$$

P_4 , deadlock

12. Draw the resource allocation graph for the following scenario:

System has three processes P_1, P_2, P_3 and has three resources R_1, R_2, R_3 (each of different type)

Event	Action
1	P_1 requests and is allocated R_1
2	P_1 requests and is allocated R_2
3	P_2 requests R_1
4	P_3 requests and is allocated R_3
5	P_1 releases R_1 , which is allocated to P_2
6	P_3 requests R_2
7	P_1 releases R_2 , which is allocated to P_3

13. Consider a system consisting of four processes and a single resource. The current state of the claim and allocation matrices are:

C	3	A	1
	2		1
	9		3
	7		2

What is the minimum number of units of the resource needed to be available for this state to be safe?

	C	A	N
P1	3	1	2
P2	2	1	1
P3	9	3	6
P4	7	2	5

Minimum number of resource needed to be available, $V = 1$.

Give to P_2 , which will complete and releases R , so $V = 2$.

Give to $P_1 \Rightarrow P_1$ completes, $V = 3$.

Not enough to give to P_5 (which need 5 units),

$\Rightarrow$ so need additional 2 units.

Therefore, the minimum units of resource needed to be available,

$$V = 1 + 2 = 3.$$

14. A resource allocation graph is a directed graph showing the states of resources and processes in a system. Based on the information given below, build the resource allocation graph and the corresponding wait-for graph.

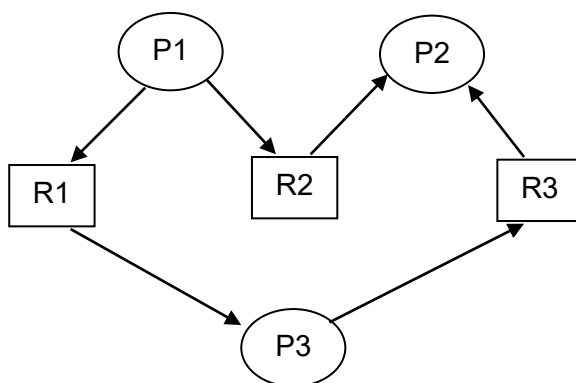
Graf peruntukan sumber adalah graf terarah yang menunjukkan keadaan sumber-sumber dan proses-proses sesuatu sistem. Berdasarkan maklumat di bawah, bina graf peruntukan sumber dan juga graf tunggu yang setara.

(Process) $P = \{P1, P2, P3\}$, (Resource) $R = \{R1, R2, R3\}$

(Edges) $E = \{P1 \rightarrow R1, P1 \rightarrow R2, P3 \rightarrow R3, R1 \rightarrow P3, R2 \rightarrow P2, R3 \rightarrow P2\}$

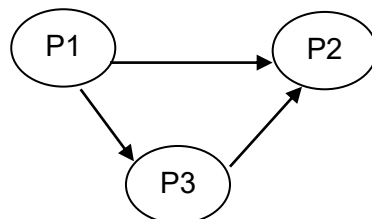
- a) Graf peruntukan sumber (*Resource allocation graph*):

Answer:



- b) Graf tunggu (*Wait-for graph*):

Answer:



- c) Based on the graph drawn, is the system in a deadlock state? Explain your answer.

Berdasarkan graf yang dilukis, adakah sistem berada di dalam keadaan kebuntuan? Terangkan jawapan anda.

Answer:

No.

Since P2 does not wait for any other processes, it can/will finish, and release the resources it holds, and P1 and P3's request can be fulfilled.

